GATE

Computer Science & Information Technology

Computer Networks

Error Control methods

DPP: 1

Q1 A building running CSMA-CD protocol has a bandwidth of 512 Mbps and a distance of 2 kilometres, and then the minimum data size is determined in order to detect a collision. Assume that the signal speed is 2,00,000 km/s

(A) 1000 bytes (B) 1250 bytes
(C) 1280 bytes (D) 1024 bytes

Q2 A transmission channel operates at 10 kbps bandwidth. Distance between two Stations in the channel are 2 km. Assume that the speed of propagation of a bit is 200 m/sec. What is the frame length if stop-and-wait ARQ is used with 50% efficiency?

(A) 25 KB (B) 50 KB
(C) 100 KB (D) 200 KB

Q3 Suppose a 10 Mbps microwave link exists between a geostationary satellite and its base station on Earth. The satellite takes a digital photo every minute and sends it to the base station. Assume a propagation speed of 2.4×10^8 meters/sec. Let x denote the size of the photo. What is the minimum value of x for the microwave link to be continuously transmitting? Assume a geostationary satellite is 36,000 kilometres away from the earth's surface.

(A) 6 Mbits (B) 4 Mbits
(C) 3 Mbits (D) 5 Mbits

Q4 Consider a 100 Mbps ethernet with a maximum distance of 400 meters. What is the smallest packet length for which you can achieve an efficiency of 80%? Assume velocity is equal to the speed of light.

(A) 2022 bits (B) 1066 bits
(C) 1520 bits (D) None

Q5 If the transmission time is 1 ms and the propagation time is 1.5 ms, what is the efficiency of stop and wait protocol?

(A) 0.25 (B) 0.05
(C) 0.30 (D) 0.50

Q6 In the Go-Back-N protocol, if the maximum window size is 127, what is the range of the sequence number?

(A) 0 to 127 (B) 0 to 128
(C) 1 to 127 (D) 1 to 128

Q7 Station 'A' uses 64 Byte packets to transmit messages to station 'B' using a sliding window protocol. The round trip delay between A and B is 80 milliseconds, and the bottleneck bandwidth on the path between 'A' and 'B' is 128 Kbps. What is the sender window size for maximum efficiency?

Q8 Consider the following message M=1010001011. The cyclic redundancy check(CRC) for this message using the divisor polynomial $x^5 + x^3 + x^2 + 1$ is:

(A) 01110 (B) 01011
(C) 10110 (D) 01101

Q9 Consider a machine M1 that wants to send a huge file to machine M2 over a TCP connection.

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Suppose the sequence number for a segment in this connection is S, then, the subsequent segment's sequence number will necessarily be S + 1.

(A) False
(B) Depends on implementation
(C) True
(D) None

Q10 Suppose there is exactly one packet switch between a sending host and a receiving host. The transmission rates between the sending host and the switch and between the switch and the receiving host are R1 and R2, respectively. Assuming that the switch uses store-and-forward packet switching, what is the total end-to-end delay in sending a packet of length L?
(Ignore queuing, propagation delay, and processing delay.)

(A) 2L/(R1 + R2)
(B) L/R1+L/R2.
(C) 2L/R1*R2.
(D) L*R1 + L*R2.

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Answer Key

Q1	(C)	Q6	(A)
Q2	(A)	Q7	21~21
Q3	(C)	Q8	(D)
Q4	(B)	Q9	(A)
Q5	(A)	Q10	(B)

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Hints & Solutions

Q1 Text Solution:

Bandwidth B = 512 Mbps

= 512 × 10^6 bits/sec

Distance d = 2 km

Speed of signal V = 2,00,000 km/s = 2 × 10^5 km/s

For CSMA - CD, to detect a collision, Ttrans ⇒ 2 * Tprop

Propagation delay Tprop = Distance/Speed of signal

= 2 km/(2 × 10^5 km/sec)

= 10^{-5} sec

Transmission delay Ttrans = Size of data / Bandwidth

= L/ (512 × 10^6 (bits/sec.)

Since, Ttrans ⇒ 2*Tprop

L = 2 × 10^5 sec × 512 × 10^6 bits/sec = 10240 bits = 1280 bytes

Q2 Text Solution:

Efficiency = 1 / (1 + 2a) ,

Where a = Tp / Tt

0.5 = 1 / (1 + 2a) ⇒ a = 1/2

Therefore, (Tp) / Tt = 1/2

⇒ (d / v) / (L / B) = 1/2

L= 2 × (2 ×10^3)/200 × 10 × 10^3

⇒ L = 25 KB

Q3 Text Solution:

To send continuously, the transmission time should be equal to the propagation time.

Tt = 2 × Tp

Length of packet / Bandwidth = 2 × distance / velocity

Length of packet = (2 × distance × Bandwidth) / velocity

= (2 × 36000 × 10^3 × 10 × 10^6)/ (2.4 × 10^8)

= 3 Mbits

Q4 Text Solution:

Given:

B = 100 Mbps, D = 400 meter, V = 3 × 10^8 m/s

⇒ T propagation = D/V = (4/3) × 10^{-6} sec

⇒ T transmission = L/B = L/10^8 sec

⇒ a = T propagation / T transmission = (4/3) × 10^{-6}/ (L/10^8)

Efficiency = 1/(1 + 2a) = 80/100, where a = T propagation/T

transmission

⇒ L/ (L + 800) = 4/5

⇒ L = 1066 bits

Q5 Text Solution:

Efficiency formula is 1/1 + 2a, a = Tp/Tt

1/1 + 2 × 1.5 = 1/4 = 0.25

Q6 Text Solution:

Sender window size is 127 (i.e. a sender window will have numbers from 0 to 126).

Now,

The sequence number required ⇒ Sender window size(127) + receiver window size(1) ⇒ 128.

Therefore, the range of sequence numbers is 0 to 127

Q7 Text Solution:

Tt = L/B = 64 Bytes / 128 Kbitps = 64 × 8 bits / 128 × 10^3 bitps = 4 × 10^{-3} sec = 4 msec

Round trip delay = 80 msec

2Tp = 80 msec

Tp = 40 msec

Optimal window size = 1 +2 × a (where a= Tp/Tt)

= 1 + 2 × 40/4 ⇒ 21

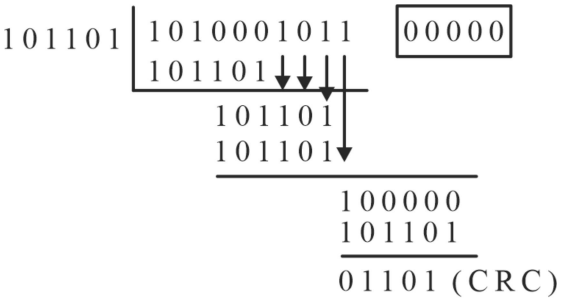
Q8 Text Solution:

Given message M = 1010001011

Divisor $x^5 + x^3 + x^2 + 1$ = 101101 Perform exclusive

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OR or MOD 2 division



Q9 **Text Solution:**

If the sequence number of the segment is S and its size is Z, Then the next sequence number will be S+Z. So, the subsequent segments can be S+1 or more depending on the size of the segment we have sent.

Q10 **Text Solution:**

At time t0, the sending host begins to transmit. At time t1= L/R1, the sending host completes transmission, and the entire packet is received at the router (no propagation delay).

Because the router has the entire packet at time t1, it can begin to transmit the packet to the receiving host at time tl. At time t2 = t1 + L/R2, the router completes transmission and the entire packet is received at the receiving host (again, no propagation delay).

Thus, the end-to-end delay is L/R1 + L/R2.

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